

Foundations of Differential privacy machine learning

[Differential privacy machine learning]

1.0 Content Block

The Laplace mechanism adds Laplace noise (i.e. noise from the Laplace distribution, which can be expressed by probability density function $\text{noise}(y) \propto \exp(-|y|/\lambda)$, which has mean zero and standard deviation $\sqrt{2}\lambda$). Now in our case we define the output function of A as a real valued function (called as the transcript output by A) as $T_A(x) = f(x) + Y$ where $Y \sim \text{Lap}(\lambda)$ and f is the original real valued query/function we planned to execute on the database. Now clearly $T_A(x)$ can be considered to be a continuous random variable, where

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Data utility and accuracy. The main concern with differential privacy is the trade-off between data utility and individual privacy. If the privacy loss parameter is set to favor utility, the privacy benefits are lowered (less “noise” is injected into the system); if the privacy loss parameter is set to favor heavy privacy, the accuracy and utility of the dataset are lowered (more “noise” is injected into the system). It is important for policymakers to consider the trade-offs posed by differential privacy in order to help set appropriate best practices and standards around the use of this privacy preserving practice, especially considering the diversity in organizational use cases. It is worth noting, though, that decreased accuracy and utility is a common issue among all statistical disclosure limitation methods and is not unique to differential privacy. What is unique, however, is how policymakers, researchers, and implementers can consider mitigating against the risks presented through this trade-off. Data privacy and security. Differential privacy provides a quantified measure of privacy loss and an upper bound and allows curators to choose the explicit trade-off between privacy and accuracy. It is robust to still unknown privacy attacks. However, it encourages greater data sharing, which if done poorly, increases privacy risk. Differential privacy implies that privacy is protected, but this depends very much on the privacy loss parameter chosen and may instead lead to a false sense of security. Finally, though it is robust against unforeseen future privacy attacks, a countermeasure may be devised that we cannot predict.

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ϵ -differentially private mechanisms Since differential privacy is a probabilistic concept, any differentially private mechanism is necessarily randomized. Some of these, like the Laplace mechanism, described below, rely on adding controlled noise to the function that we want to compute. Others, like the exponential mechanism and posterior sampling sample from a problem-dependent family of distributions instead. An important definition with respect to ϵ -differentially private mechanisms is sensitivity. Let d be a positive integer, D be a collection of datasets, and $f : D \rightarrow \mathbb{R}^d$ be a function. One definition of the sensitivity of a function, denoted Δf , can be defined by: $\Delta f = \max_{D_1, D_2} \|f(D_1) - f(D_2)\|_1$, where the maximum is over all pairs of datasets D_1 and D_2 in D differing in at most one element and $\|\cdot\|_1$ denotes the L1 norm. In the example of the medical database below, if we consider f to be the function Q_i , then the sensitivity of the function is one, since changing any one of the entries in the database causes the output of the function to change by either zero or one. This can be generalized to

other metric spaces (measures of distance), and must be to make certain differentially private algorithms work, including adding noise from the Gaussian distribution (which requires the L2 norm) instead of the Laplace distribution. There are techniques (which are described below) using which we can create a differentially private algorithm for functions, with parameters that vary depending on their sensitivity.

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Now suppose a malicious user (often termed an adversary) wants to find whether Chandler has diabetes or not. Suppose he also knows in which row of the database Chandler resides. Now suppose the adversary is only allowed to use a particular form of query Q_i that returns the partial sum of the first i rows of column X in the database. In order to find Chandler's diabetes status the adversary executes $Q_5(D_1)$ and $Q_4(D_1)$, then computes their difference. In this example, $Q_5(D_1) = 3$ and $Q_4(D_1) = 2$, so their difference is 1. This indicates that the "Has Diabetes" field in Chandler's row must be 1. This example highlights how individual information can be compromised even without explicitly querying for the information of a specific individual. Continuing this example, if we construct D_2 by replacing (Chandler, 1) with (Chandler, 0) then this malicious adversary will be able to distinguish D_2 from D_1 by computing $Q_5 - Q_4$ for each dataset. If the adversary were required to receive the values Q_i via an ϵ -differentially private algorithm, for a sufficiently small ϵ , then he or she would be unable to distinguish between the two datasets.

5.0 Content Block

Differential privacy (DP) is a mathematically rigorous framework for releasing statistical information about datasets while protecting the privacy of individual data subjects. It enables a data holder to share aggregate patterns of the group while limiting information that is leaked about specific individuals. This is done by injecting carefully calibrated noise into statistical computations such that the utility of the statistic is preserved while provably limiting what can be inferred about any individual in the dataset. Another way to describe differential privacy is as a constraint on the algorithms used to publish aggregate information about a statistical database which limits the disclosure of private information of records in the database. For example, differentially private algorithms are used by some government agencies to publish demographic information or other statistical aggregates while ensuring confidentiality of survey responses, and by companies to collect information about user behavior while controlling what is visible even to internal analysts. Roughly, an algorithm is differentially private if an observer seeing its output cannot tell whether a particular individual's information was used in the computation. Differential privacy is often discussed in the context of identifying individuals whose information may be in a database. Although it does not directly refer to identification and reidentification attacks, differentially private algorithms provably resist such attacks.